JEETENDRA GUPTA

Senior Python Backend, API & Cloud Developer

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PROFESSIONAL SUMMARY

*Senior Python Backend, API & Cloud Developer with **9 years of experience** building production-grade backend systems across **fintech, cloud infrastructure, environmental technology, e-commerce, and power**. Skilled at turning complex business requirements into practical technical solutions and building systems that stay reliable under real-world load.*

*Experience spans **backend API development, cloud automation, infrastructure monitoring, and large-scale data platforms**. Built a monitoring platform for **1,800+ Paytm servers**, automated **GCP backup pipelines for 1,500+ virtual machines**, and designed APIs for **carbon credit data platforms used by Indian government ministries** — work where reliability, scalability, and data accuracy were non-negotiable.*

*Core stack: **Python, Flask, FastAPI, REST APIs, PostgreSQL, Docker, GCP, and AWS**. Focused on clean, maintainable, testable software that teams can confidently build on, with a strong emphasis on **reliability, automation, and long-term maintainability**.*

TECHNICAL SKILLS

LANGUAGES

Python | JavaScript | Shell / Bash | SQL

FRAMEWORKS & LIBRARIES

Flask | FastAPI | SQLAlchemy | Pydantic | Behave (BDD) | Jinja2 | Pandas

BACKEND & APIS

REST API | GraphQL | JWT | OAuth2 | RBAC | Gunicorn | Uvicorn | Swagger / OpenAPI

DATABASES

PostgreSQL | MySQL | MariaDB | MongoDB | Redis | SQLite3 | DynamoDB | SAP HANA | PostGIS

CLOUD

AWS EC2 / S3 / Lambda | AWS RDS / API Gateway | GCP Compute Engine | GCP Cloud Storage | GCP Cloud Functions | Azure Blob Storage | Azure REST APIs | Azure HANA Backup | STACKIT S3 Object Storage | STACKIT IMDS Auth | boto3 / botocore | RefreshableCredentials | IBM Cloud Object Storage | IBM VPC Metadata API | IBM Power Systems SDK | trusted-profile auth

DEVOPS & CI/CD

Docker | Kubernetes | Ansible | Jenkins | GitHub Actions | Nagios | OpenNMS | Nginx

TESTING

Pytest | Unittest | Behave BDD | Postman | Integration Testing

TOOLS & PLATFORMS

Git / GitHub | SVN | JIRA | PyCharm | VS Code | Linux RHEL | CentOS | Ubuntu | Chart.js | Leaflet

PROFESSIONAL EXPERIENCE

Senior Python Backend & Cloud Developer

NTT Data Business Solutions , Client: SAP Labs India Pvt. Ltd. — **Jul 2024 – Present**

- Automated backup and restore for 1,500+ GCP VMs with Python (subprocess module and GCP CLI), replacing a manual, fragile process with an unattended nightly run and cutting manual overhead by roughly 70%.
- Built a full BDD test suite for the Snappy Agent using Python Behave, taking backup coverage from near zero to 90%+ and reducing critical job failures from 15% to 0.2% within a single quarter.
- Owned the GCP VM snapshot lifecycle across 20+ projects, automating creation, retention, and deletion through Python scripts using GCP SDK and gcloud CLI with no manual intervention.
- Tightened over-broad storage retention policies, saving roughly $18K per year in GCP storage costs (a 25% reduction) with no impact on recovery capability.
- Built FastAPI internal services consumed daily by 10+ engineering teams, with clean documentation and predictable behavior that let teams integrate without back-and-forth.
- Mentor junior developers through code reviews focused on the reasoning behind decisions; several have gone on to lead their own features independently.

Senior Python Backend Developer

Iora Ecological Trust , Client: Iora Ecological Solutions Pvt. Ltd. — **Oct 2023 – Jun 2024**

- Built the SDG Monitoring Dashboard and its full backend for the Assam state government, giving planners a real-time view of development progress across health, education, and environment.
- Served as SDG Technical Consultant for Jammu and Kashmir, working directly with government stakeholders to scope and design a dashboard surfacing indicators that informed policy decisions.
- Developed the Digital Carbon Toolbox in Python, Flask, and PostgreSQL to calculate carbon emissions for tree projects using Wildlife of India scientific datasets, used directly by carbon teams for field reporting.
- Built the DRC Visualisation Portal for the Democratic Republic of Congo, rendering environmental layers — watersheds, habitats, forests, and land use — through Leaflet and Chart.js on a Python and Flask backend.
- Led technical R&D for India's Ministry of Parivahan Sewa carbon credit platform, enabling large enterprises including Reliance and Tata to calculate and report carbon credits through the GCP-Portal.
- Contributed to the SLMC biodiversity project, building remote sensing visualisations and mapping dashboards used by field teams for daily ecological decision-making.

Python Backend Developer

Jindal India Thermal Power Ltd. , Client: Amala Earth Private Limited **Feb 2023 – Oct 2023**

- Designed and built RESTful APIs in Flask and FastAPI, including a SAP HANA Data Layer API with role-based access control that restricted each user group to exactly the data it was permitted to query.
- Set up automated CI/CD pipelines that cut deployment time by 35%, replacing manual, inconsistent deployments that had caused frequent release-time issues.
- Used Ansible for preventive maintenance, system upgrades, and server configuration management, turning stressful production deployments into a routine, predictable process.
- Built reporting modules in HTML, CSS, and JavaScript for plant engineers, prioritising clarity for non-developer end users.
- Ran code reviews, mentored junior team members, and drove consistent API design standards that made cross-team integrations noticeably smoother.

Senior Software Developer

Infinite Computer Solutions , Client: Nokia Solutions and Networks India Private Limited **Jan 2022 – Feb 2023**

- Built custom Python reporting modules for Nokia IMPACT 21/22 that let the client run their own SQL queries and view results in real time, eliminating the previous ticket-and-wait cycle for custom reports.
- Deployed Nokia IMPACT 21/22, MariaDB, and Nagios across multiple hosts using Ansible Playbooks, reducing system downtime by 20% through consistent, repeatable configurations.
- Developed monitoring tools with automatic email escalation on threshold breaches, freeing engineers from constant dashboard watching and reducing mean time to detect (MTTD) by 40%.
- Onboarded several new developers with proper documentation and hands-on ramp-up support through their early weeks.
- Recognised with Best Performer of the Month in September 2022 and the Spark Award for Q2 2022, both for outstanding delivery and client satisfaction.

Software Developer

One97 Communications Ltd. (Paytm) , Client: Paytm Payment Services Limited **May 2016 – Jan 2022**

- Built EGL, an internal server monitoring panel tracking 1,800+ production servers using Nagios and OpenNMS, which became a daily-relied-upon tool for the operations team over nearly six years as the platform scaled into one of India's largest payment companies.
- Automated hardware inventory tracking and real-time critical event alerting across the server fleet, contributing directly to a 30% reduction in production downtime over the following year.
- Developed internal Flask REST APIs for operational dashboards used daily by 200+ engineers and operations staff, kept simple and well documented for the many teams that depended on them.
- Led a codebase modernisation effort, upgrading several legacy Python services to current standards and reducing technical debt across the team.
- Partnered closely with product, UAT, and development teams on scoping and delivery, translating vague requirements into buildable, maintainable solutions.

Network Support Engineer

Accel Frontline Services Ltd. , Client: Timex Group India Limited — **Jul 2014 – May 2016**

- Handled installation and configuration of network hubs, web servers, and storage servers, maintaining SLA compliance above 95%.
- Wrote Python scripts to automate repetitive reporting tasks, freeing up meaningful team time and sparking a lasting interest in automation.
- Prepared thorough post-mortems for network failures, establishing a habit of documenting root causes and fixes.
- Diagnosed and resolved issues across hardware, software, and connectivity, building a strong bottom-up understanding of systems.

KEY PROJECTS

SAP Data Layer API Development – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, FastAPI, SAP HANA

- Designed role-based APIs over SAP HANA that restricted each user group to authorised data, enforcing access control at the API layer rather than the database alone for stronger protection.
- Tuned queries for large-scale datasets, noticeably reducing latency on high-volume data access used daily by multiple SAP engineering teams.
- Kept the API design clean and Swagger documentation thorough, enabling teams to integrate without repeated support.

Automated GCP VM Snapshot Management – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, GCP SDK, gcloud CLI

- Automated VM snapshot creation, retention enforcement, and deletion across the entire GCP fleet in Python, removing all manual triggering.
- Tightened previously over-broad retention policies, cutting storage costs by roughly 25% with no impact on recovery capability.
- Added error handling and alerting so failed snapshot jobs are flagged immediately rather than discovered hours later during an incident.

Snappy Agent-Based File System Backup Testing – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, Behave (BDD), CI/CD

- Built a full BDD test suite in Python Behave covering backup, restore, and edge-case scenarios end to end, starting from essentially zero coverage.
- Authored feature files and step definitions broad enough to catch regressions in core backup paths automatically before they reach production.
- Cut the manual testing cycle significantly by moving hours of repetitive scenario testing into CI on every push.

Snappy Detect (Cloud Provider Detection System) – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, Shell, JSON, Terminal & Crontab

- Built automatic cloud provider detection by analyzing file paths and installed CLI tools, with OS version detection to trigger OS-specific commands.
- Added detection of installed databases, services, and RPMs, presenting results in a readable terminal table with JSON export.
- Improved operational efficiency across multi-cloud environments.

STACKIT Cloud HANA Backup, Restore & Log Monitoring – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, boto3, botocore, STACKIT IMDS Auth, RefreshableCredentials, S3 Object Storage, SAP HANA, ne.db-hdbbackint, Behave BDD

- Delivered the complete HANA backint backup, restore, and listing solution for STACKIT cloud from scratch, covering full, differential, incremental, and log backup types across both Scale-Up and Scale-Out HANA systems.
- Implemented STACKIT IMDS-based service account authentication with botocore RefreshableCredentials for automatic token refresh during long-running backups, resolving silent mid-backup failures on jobs running over an hour.
- Diagnosed and fixed multiple backup-listing bugs, including incorrect S3 key prefixes, a PartSizeMB value that exceeded S3 multipart limits, and a vendor issue in ne.db-hdbbackint v0.8.2 that required a --backintVersion workaround.
- Integrated HANA log backup reporting and monitoring into Octobus and TIC eventing, and registered STACKIT detector classes in snappy-detect for discoverability by the agent framework.

IBM Power Systems Trusted-Profile Auth & NFS Backup Enhancement – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, IBM Cloud Metadata API, IBM VPC API, IBM Power Systems SDK, platform.machine(), snappy-agent CLI

- Fixed trusted-profile authentication for IBM Power Systems by auto-detecting server architecture at runtime with platform.machine() and routing to the correct metadata endpoint, working transparently across 1,800+ existing deployments with zero configuration file changes.
- Added an --include-nfs option to snappy-agent for IBM and ACS providers, enabling ad-hoc NFS filesystem backups from the CLI, matching the capability GCP users already had.

GCP NFS Backup Monitoring & Event Reporting – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, GCP GCFS, Octobus, TIC, APScheduler, File-based Caching

- Moved GCP GCFS NFS backup monitoring from the FS backup script layer into the snappy-backup feature framework so it emits correctly typed NFS events to Octobus and TIC instead of mislabeled filesystem events.
- Added a 12-hour result cache to prevent duplicate monitoring runs and a 24-hour notify gate limiting Octobus to one success notification per day, while keeping failure notifications on every cycle so nothing is missed.
- Implemented an APScheduler cron job inside snappy-agent that runs NFS monitoring on an independent schedule, decoupled from FS backup execution and controlled by a single config flag with no agent restart required.

GCP File System Backup & Restore Automation – NTT Data Business Solutions (Client: SAP Labs India Pvt. Ltd.)

Python, GCP CLI, Subprocess, Cloud Storage

- Rewrote a manual, fragile GCP backup and restore process in Python using subprocess and GCP CLI calls, now running unattended on schedule with no manual involvement.
- Integrated GCP Cloud Storage to make disaster recovery workflows reliable and repeatable across the environment rather than dependent on the on-call engineer.
- Replaced a set of scripts that required constant babysitting, noticeably improving operational efficiency.

SDG Monitoring Dashboard & Backend Services – Iora Ecological Trust (Client: Assam State Government)

Python, Flask, PostgreSQL, Leaflet, Chart.js

- Built the full backend for the Assam Government's SDG tracking platform, giving planners live progress across health, education, and environment indicators without waiting for manually compiled reports.
- Developed the REST APIs and data aggregation layer, consolidating multiple government data sources into a single consistent view for state-level planning decisions.
- Delivered complex data to non-technical decision makers in a form they could act on directly.

Digital Carbon Toolbox (Carbon Emission Calculator) – Iora Ecological Trust (Client: India's Ministry of Parivahan Sewa)

Python, Flask, PostgreSQL, AWS

- Built backend services that calculate carbon emissions for tree projects using Wildlife of India scientific datasets, with calculation logic accurate enough to withstand government scrutiny.
- Integrated emission data into existing system APIs so carbon teams could generate reports without switching tools or running separate processes.
- Designed the output and workflow for non-technical carbon field workers, requiring no training to use.

SDG Technical Consultant (Jammu & Kashmir State Government) – Iora Ecological Trust

Azure, SQL Server, Python, Flask

- Advised government stakeholders on system architecture and data visualization strategies, and designed the SDG Dashboard to track environmental, social, and economic indicators with real-time data feeds.
- Built a multi-cloud architecture using Azure, SQL Server, and Flask for scalable IaC deployment.
- Enabled tracking of 15+ SDG indicators and reduced infrastructure costs by 25% through optimization.

GCP-Portal for Green Credit Program – Iora Ecological Trust (Client: India's Ministry of Parivahan Sewa)

Python, Flask, PostgreSQL, GCP

- Led technical assessments and contributed to portal development, enabling carbon credit calculations for large enterprises including Reliance and Tata.
- Streamlined carbon footprint monitoring and reporting, building a robust backend supporting enterprise-scale calculations.

Green Credit Program Portal (Azure) – Iora Ecological Trust

Azure, Microsoft SQL, FastAPI

- Built a cloud portal for enterprise carbon credit calculations, provisioning the Carbon Credit Database using Azure Microsoft SQL.
- Implemented JWT and RBAC security throughout.

Remote Sensing & Mapping Services (Bioeconomy Project, DRC) – Iora Ecological Trust (Client: Iora Ecological Solutions Pvt. Ltd.)

Python, Flask, PostGIS, Leaflet, Chart.js

- Set up PostGIS-backed GIS servers to manage geospatial data for the DRC bioeconomy project, covering land use, watershed boundaries, habitats, and forest cover across large geographic areas.
- Built Leaflet-based mapping dashboards for field teams and analysts to visualise ecosystem metrics interactively instead of working from raw data exports.
- Fed the resulting insights directly into biodiversity and conservation planning for one of the world's largest tropical forest regions.

DRC Data Visualization & GIS Portal – Iora Ecological Trust

Flask, PostGIS, Leaflet, Chart.js

- Built GIS dashboards with map layers and analytics, visualizing watersheds, forests, and land-use data.
- Enabled data-driven ecological decisions for conservation planning.

SLMC Project — Landscape Management & Conservation – Iora Ecological Trust

Python, Leaflet, Chart.js, Flask

- Enhanced ecological landscape management for biodiversity and climate change mitigation.
- Developed remote sensing data visualizations, mapping dashboards, and reporting tools.
- Supported decision-making and project tracking using Leaflet and Chart.js.

Flower and Butterfly GIS Data Collector – Iora Ecological Trust

KoboToolbox, ODK, Python, Pandas, PostgreSQL, Mobile Data Collection

- Built mobile data collection forms using KoboToolbox and ODK for offline field capture, with automated data sync pipelines to a centralized PostgreSQL database.
- Developed Python/Pandas scripts for data cleaning and species analysis.
- Enabled GIS data collection from 250+ field researchers and reduced errors by 60%.

Amala Earth Finance Module – Jindal India Thermal Power Ltd. (Client: Amala Earth Private Limited)

Python, Pandas, Flask, PostgreSQL

- Automated seller payout calculations from delivery reports, replacing a slow, error-prone manual month-end process with consistent, repeatable results.
- Integrated the finance logic into existing system APIs so payout data flows through the same pipeline as everything else, with no separate process to maintain.
- Used Pandas to generate Excel reconciliation reports the finance team could hand directly to auditors without reformatting.

Shopify & Unicommerce Marketplace Integration – Jindal India Thermal Power Ltd. (Client: Amala Earth Private Limited)

Python, FastAPI, JWT, PostgreSQL, REST APIs

- Built the backend API layer that keeps product, order, inventory, and shipment data in sync between Shopify and Unicommerce in real time, replacing a process that had needed regular manual intervention to stay consistent.
- Built in JWT-based authentication throughout; the system has run reliably with minimal maintenance since launch.

Nokia IMPACT 21/22 Custom Reporting & Monitoring – Infinite Computer Solutions (Client: Nokia Solutions and Networks India Private Limited)

Python, MariaDB, Oracle, Ansible, Nagios

- Built custom Python reporting modules for Nokia IMPACT 21/22 enabling the client to run their own SQL queries with real-time results, eliminating the previous multi-day manual report request cycle.
- Automated system monitoring with Python scripts and Nagios integrations, sustaining high availability without constant dashboard watching.
- Deployed MariaDB and Oracle instances across multiple hosts using Ansible Playbooks, making rollouts repeatable and cutting setup time significantly versus manual installation.
- Optimised slow queries and tuned system performance proactively, improving reporting speed and reducing database host load during peak usage.

OpenNMS Push-Agent & Agent-Less Monitoring Tool – One97 Communications Ltd. (Client: Paytm Payment Services Limited)

Python, Flask, XML, OpenNMS, PostgreSQL, Psutil, Google Charts

- Built two complementary monitoring tools: a push-agent collecting CPU, memory, disk, and network metrics every 5 minutes and pushing them to OpenNMS in XML, and an agent-less version doing the same remotely with nothing installed on target servers.
- Integrated OpenNMS to convert incoming XML into RRD time-series metrics, enabling consistent long-term performance trending across the server fleet.
- Built a Flask dashboard with Google Charts for an interactive performance view without querying OpenNMS directly.

Docker Duck, Container Health Monitor – One97 Communications Ltd. (Client: Paytm Payment Services Limited)

Python, Shell, PostgreSQL, Google Graph API, Docker

- Built a container health and performance monitor that stores metrics in PostgreSQL, visualises them through Google Graph API, and sends email alerts on anomalies, closing a gap where container issues were only caught after they broke.
- Shifted container monitoring from reactive to proactive, alerting the team before failures reached users.

PayTM Server Monitoring Panel – One97 Communications Ltd. (Client: Paytm Payment Services Limited)

Python, Flask, Nagios, OpenNMS, PostgreSQL

- Built a single trusted monitoring view for 1,800 production servers, handling automated hardware inventory, real-time health checks, and critical event alerting.
- Established the panel as a core internal tool; the 30% drop in production downtime that followed came directly from teams catching issues before they escalated into incidents.

Automated MIS Reporting System – Accel Frontline Services Ltd. (Client: Timex Group India Limited)

Shell, Batch and PowerShell Script, SMS API, Windows Management Instrumentation, CSV, Text Processing

- Built shell, batch, and PowerShell scripts using Windows Management Instrumentation to automatically collect system, hardware, and operational data across multiple machines, formatting it into structured MIS reports delivered to the HOD.
- Set up threshold-based alerting that fired an SMS notification via SMS API to the right person whenever a tracked metric crossed its limit, requiring no manual monitoring.
- Converted a daily manual reporting routine into a fully scheduled, zero-touch process with immediately visible time savings.
- Covered uptime, disk usage, memory utilisation, and network health, giving management a consistent daily snapshot of IT infrastructure.

EDUCATION

Post Graduate Diploma in Advanced Computing (PG-DAC)

Centre for Development of Advanced Computing (C-DAC) , Delhi, India — **2013**

Bachelor of Technology in Computer Science Engineering

Maharshi Dayanand University (MDU) , Haryana, India — **2012**

Higher Secondary, Science, Mathematics and Computer Science

Ebenezer Matriculation Higher Secondary School , Tamil Nadu, India — **2008**

CERTIFICATIONS

- **Certified Kubernetes Professional**, Infinite Certified Kubernetes Association
- **Cloud Foundations (AWS, GCP and Azure)**, Infinite Cloud Foundation
- **Red Hat Certified System Administrator**, Red Hat Enterprise Linux 7 and 8
- **DB2 Database Management and Optimisation**, IBM, 3-month programme

INDUSTRIAL TRAINING & INTERNSHIPS

- **IBM DB2:** 3-month database management & optimization training.
- **HTML & Web Fundamentals:** 6-week front-end development training (Oxford Computer Academy).
- **Java Training:** 4-week Java programming course + 6-week internship at J.K Technologies Pvt. Ltd.
- **PHP & PHP++:** 6-month training + 3-month internship at Matrix Infosys.

AWARDS & ACHIEVEMENTS

- **Best Performer of the Month, Sep 2022**, Infinite Computer Solutions, recognised for outstanding delivery and client satisfaction.
- **Spark Award, Q2 2022 (Jul to Sep)**, Infinite Computer Solutions, awarded for exceptional performance over the quarter.
- **GitHub Pull Shark Badge**, recognised for consistent high-quality open-source pull request contributions.

ADDITIONAL INFORMATION

- **Open Source:** Maintains 18+ public Python repositories on GitHub covering FastAPI, Flask, GCP automation, and geospatial tooling.
- **Industrial Training:** Java at Informatics Computer Education, PHP and PHP++ for 6 months at MultiSoft Systems, HTML and Web at Oxford Computer Academy, Java Internship at J.K Technologies, PHP Internship at Matrix Infosys.
- **Languages:** English (professional working proficiency), Hindi (native)